Supplementary Materials: Mod, Sweat, and Tears

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1 Links

GitHub Repo Link:
[URLanonymisedforblindreview]
Paraquet Data:
[URLanonymisedforblindreview]

2 Principal Component Analysis (PCA)

2.1 PCA Scree Plot

The Scree Plot visualises the cumulative explained variance by each principal component. This was used to determine the number of components retained for further analysis.

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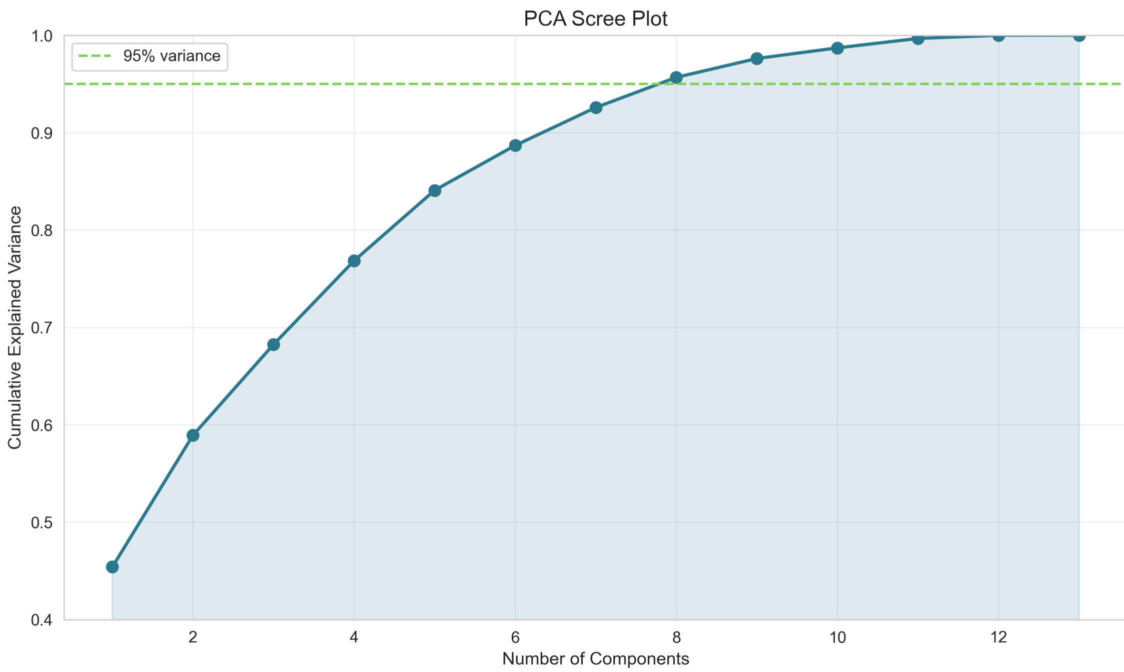


Fig. 1. PCA Scree Plot displaying cumulative explained variance used to determine the number of components retained.

2.2 PCA Loadings Heatmap

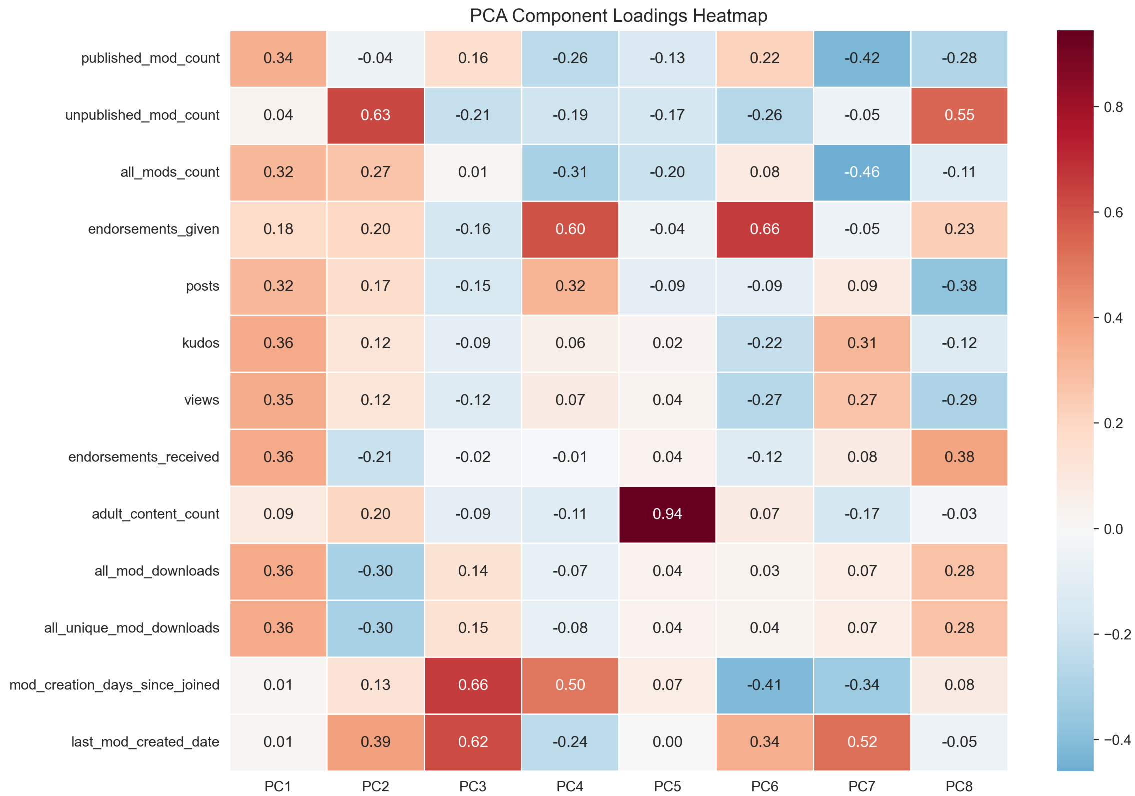


Fig. 2. Heatmap of PCA loadings illustrating feature contributions across principal components.

Table 1. Detailed PCA loadings table showing the weight of each original feature on principal components.

Feature	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
Published Mods	0.345	−0.044	0.157	−0.255	−0.132	0.220	−0.418	−0.281
Unpublished Mods	0.035	0.628	−0.214	−0.187	−0.165	−0.263	−0.053	0.552
All Mods Count	0.316	0.265	0.012	−0.306	−0.204	0.084	−0.460	−0.113
Endorsements Given	0.184	0.198	−0.162	0.600	−0.044	0.660	−0.052	0.233
Posts	0.316	0.170	−0.154	0.319	−0.088	−0.088	0.085	−0.379
Kudos	0.362	0.120	−0.090	0.060	0.023	−0.223	0.311	−0.121
Views	0.349	0.122	−0.125	0.073	0.038	−0.272	0.274	−0.287
Endorsements Received	0.362	−0.209	−0.017	−0.010	0.043	−0.115	0.078	0.380
Adult Content Count	0.089	0.204	−0.088	−0.111	0.944	0.074	−0.165	−0.025
Total Downloads	0.357	−0.305	0.137	−0.074	0.042	0.033	0.069	0.277
Unique Downloads	0.357	−0.301	0.147	−0.084	0.041	0.036	0.071	0.279
Days Since Joined	0.013	0.130	0.655	0.501	0.070	−0.411	−0.343	0.079
Last Mod Created	0.014	0.386	0.616	−0.245	0.001	0.343	0.517	−0.049

3 Devotion Score

3.1 Devotion Score Distribution

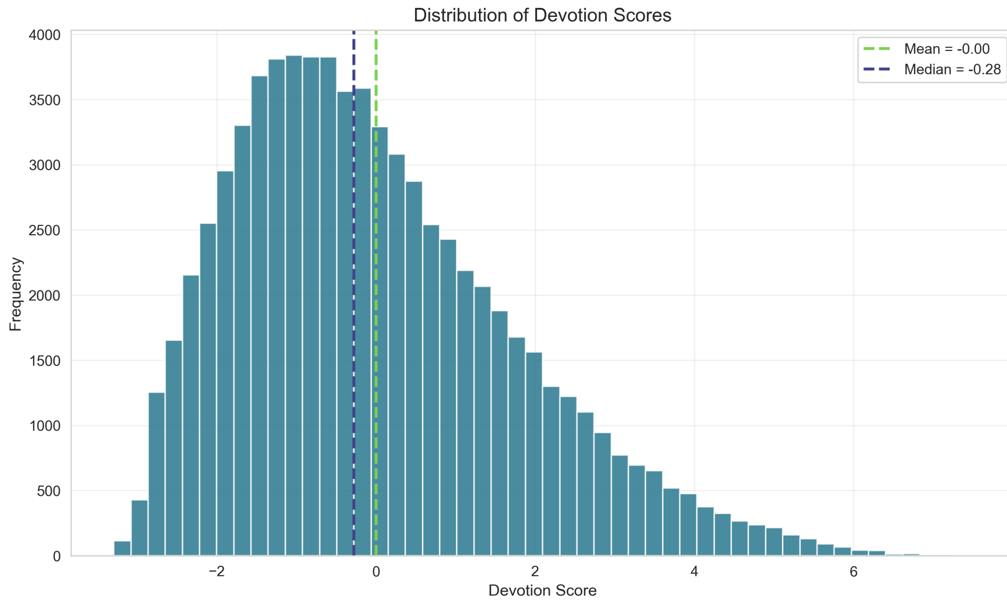


Fig. 3. Distribution of Devotion Scores across the user dataset.

3.2 Summary Statistics Table

Table 2. Summary statistics for the Devotion Score (N=73,855).

Statistic	Value
Count	73 855
Mean	-0.000
Std. deviation	1.767
Min	-3.295
25% (Q1)	-1.334
Median (Q2)	-0.280
75% (Q3)	1.069
Max	7.482
Skewness	0.739
Top 5% threshold	3.374

3.3 Feature Correlations with the Devotion Score

Table 3 reports the Spearman rank correlation between each of the thirteen behavioural features and the composite Devotion Score. Spearman's ρ is used in preference to Pearson's r because the underlying features are heavily right-skewed; rank correlation is invariant to the monotonic transformations applied during preprocessing. Community-recognition and productivity metrics (kudos, views, endorsements received, published-mod count, and downloads) correlate most strongly with the Devotion Score, whereas adult-content count, account tenure, and unpublished-mod count show only weak associations. All correlations are statistically significant at $p < .001$, which is expected given the large sample ($N = 73{,}855$).

Table 3. Spearman rank correlation between each behavioural feature and the Devotion Score, ordered by strength of association. All $p < .001$.

Feature	**Spearman ρ**
Kudos	0.832
Views	0.818
Endorsements Received	0.816
Published Mods	0.813
Unique Downloads	0.804
Total Downloads	0.803
All Mods Count	0.786
Posts	0.761
Endorsements Given	0.487
Adult Content Count	0.252
Days Since Joined	0.170
Last Mod Created	0.154
Unpublished Mods	0.139

4 Clustering Analysis

4.1 Cluster Plot

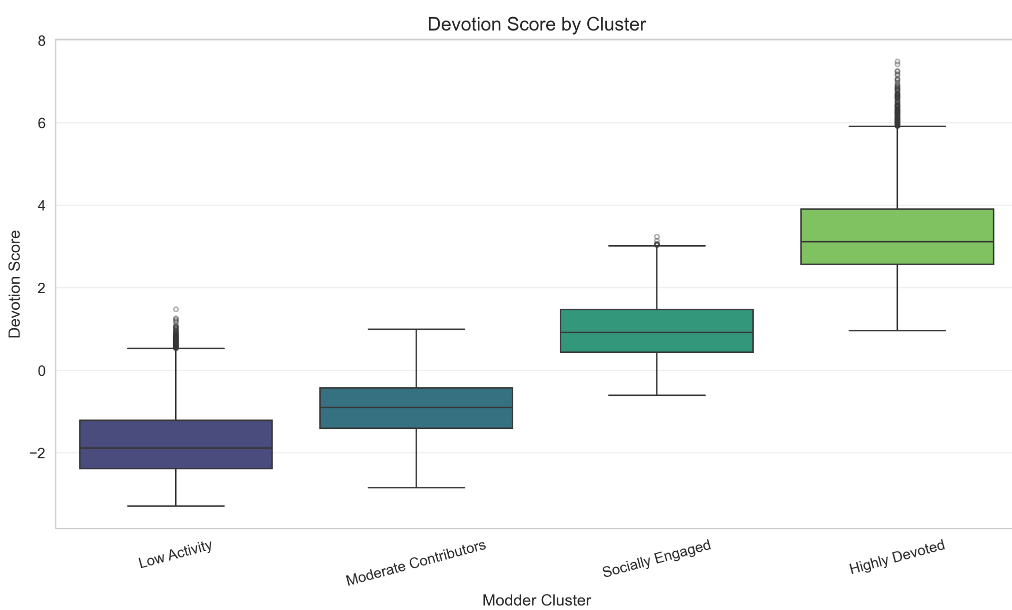


Fig. 4. Mean values of key behavioural metrics for each modder cluster, derived from K-Means clustering.

4.2 Elbow Method and Silhouette Analysis

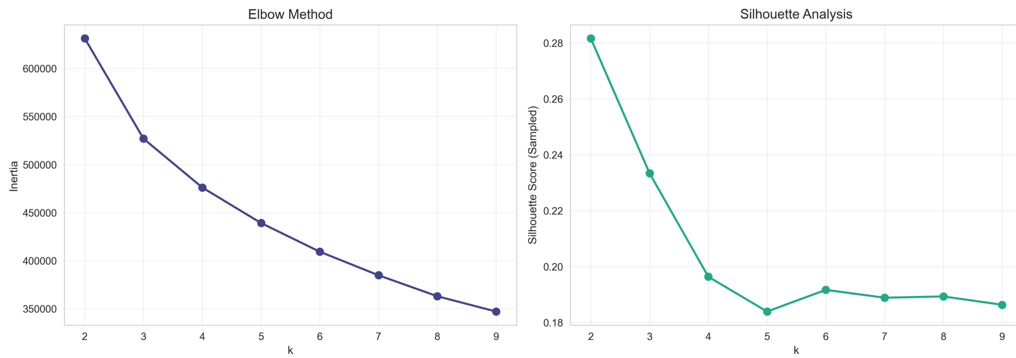


Fig. 5. Elbow method (inertia) and mean silhouette score for $k = 2$ to $k = 9$, used to select the optimal number of clusters.

5 K-Means Cluster Validation

5.1 Silhouette Score

The final clustering (K-means on PCA-reduced data) achieved a Silhouette score of 0.20, indicating moderate overlap among clusters. This result was obtained after applying data transformations to reduce skewness (logarithmic, cube root, winsorization, and Yeo–Johnson transformations), improving cluster interpretability compared to raw, untransformed data.

Table 4. Silhouette scores across clustering methods.

Method	Silhouette Score	Notes
K-Means (8 PC)	0.20	Final clustering input
K-Means (2 PC)	0.40	For interpretability only
DBSCAN	0.06	eps = 1.0, min_samples = 10
HDBSCAN	0.23	min_cluster_size = 50
GMM	0.16	Moderate separation

5.2 Data Transformations and Skewness Reduction

Table 5. Skewness before/after transformations applied to features.

Feature	Original Skew	Transformation Applied	Post-Transformation Skew
active days	0.0858	None	0.0858
recognised author	−0.3129	None	-0.3129
published_mod_count	22.7498	Yeo–Johnson	0.1804
unpublished_mod_count	36.8830	Yeo–Johnson	0.5186
contributed_mod_count	73.6131	Excluded due to skewness	–
all_mods_count	22.0369	Yeo–Johnson	0.4561
endorsements_given	9.7538	Logarithmic	0.2082
posts	32.6900	Logarithmic	0.2937
kudos	34.3876	Logarithmic	1.2837
views	3.7799	Winsorized + cube root	1.1243
donations_enabled	−0.5108	None	-0.5108
total_categories	5.2100	Yeo–Johnson	0.5123
endorsements_received	26.1154	Yeo–Johnson	0.5552
adult_content_count	26.1154	Yeo–Johnson	0.5552
all_mod_downloads	38.0076	Yeo–Johnson	-0.0191
all_unique_mod_downloads	35.4212	Yeo–Johnson	-0.0225
mod_creation_days_since_joined	1.3344	None	1.3344
Binary Features	–	None	(no change)

5.3 Cluster Averages

Table 6. Descriptive statistics by cluster (median values). K-means on first 8 PCs.

Feature	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Published Mods	0.0	1.0	3.0	11.0
Endorsements Received	0.0	35.0	232.0	2590.0
Mod Downloads	0.0	925.0	6089.0	105,238.0
Unique Downloads	0.0	668.0	4028.0	58,756.0
Comments (Posts)	5.0	6.0	49.0	299.0
Kudos Received	0.0	0.0	5.0	51.0
Devotion Score (Med)	−1.89	−0.90	0.92	3.11

5.4 Devotion Scores and Clusters

Table 7. Devotion Score distribution by cluster.

Cluster	Mean	Median	Std	Min	Max
0	−1.74	−1.89	0.83	−3.30	1.48
1	−0.93	−0.90	0.65	−2.85	0.99
2	0.97	0.92	0.67	−0.61	3.24
3	3.32	3.11	1.00	0.96	7.48

5.5 Temporal Statistics

Table 8. Temporal statistics by cluster. *Activity span* is the number of days between a user's first and last mod upload. *Active days* is the number of days between account creation (`joined`) and last platform activity (`last_active`).

Cluster	Span (Mean)	Span (Median)	Active Days (Mean)	Active Days (Median)
1 (Low Activity)	57.3	0	2846	2607
2 (Moderate)	59.2	0	3363	3456
3 (Socially Engaged)	506.2	120	3696	3845
4 (Highly Devoted)	1315.0	942	3916	4072

5.6 Feature Correlation Matrix

Presents a correlation matrix of key modder metrics. Notable patterns include a strong positive correlation between *Total Downloads* and *Unique Downloads* ($r > 0.9$), as expected. The modest correlation between *Published Mods* and *Endorsements Received* suggests that productivity alone does not guarantee community recognition.

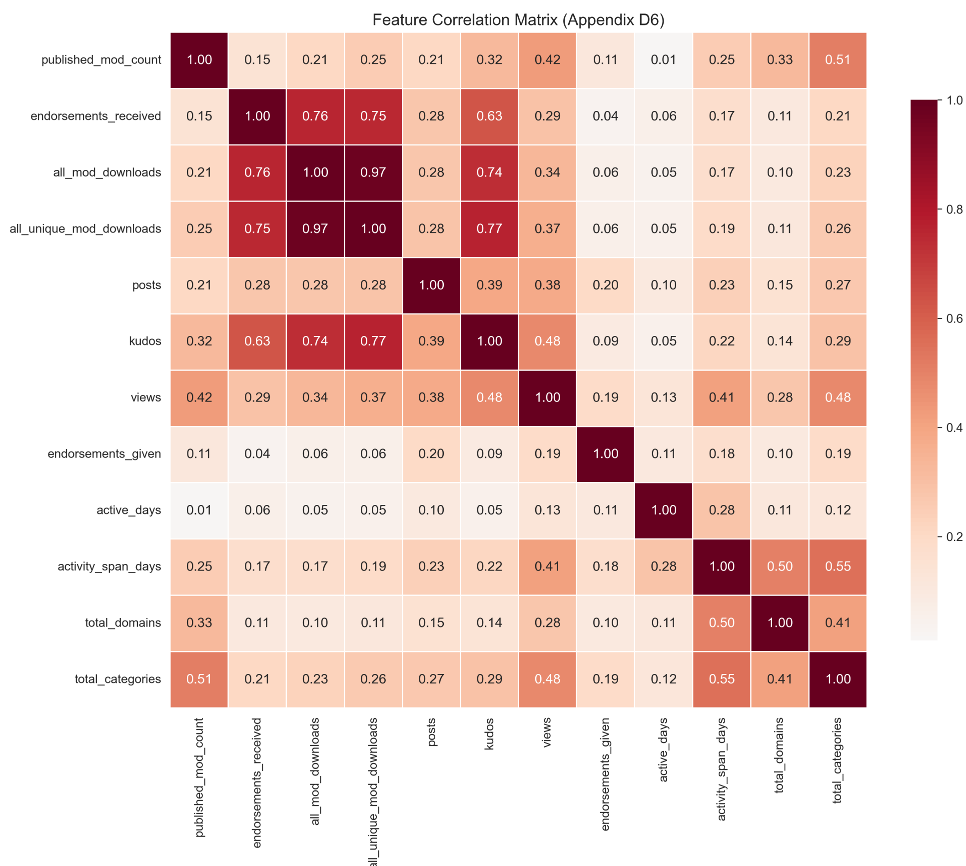


Fig. 6. Correlation matrix of key engagement and productivity metrics.

5.7 Cluster Distribution Boxplots

Visualizes the distribution of key metrics (Active Days, Endorsements Given, Forum Posts, and Total Downloads) across the four modder clusters. These boxplots illustrate the variance within each cluster, showing that the "Highly Devoted" cluster (Cluster 4) displays substantially higher median values across all metrics, supporting the interpretation of these individuals as Serious Leisure practitioners.

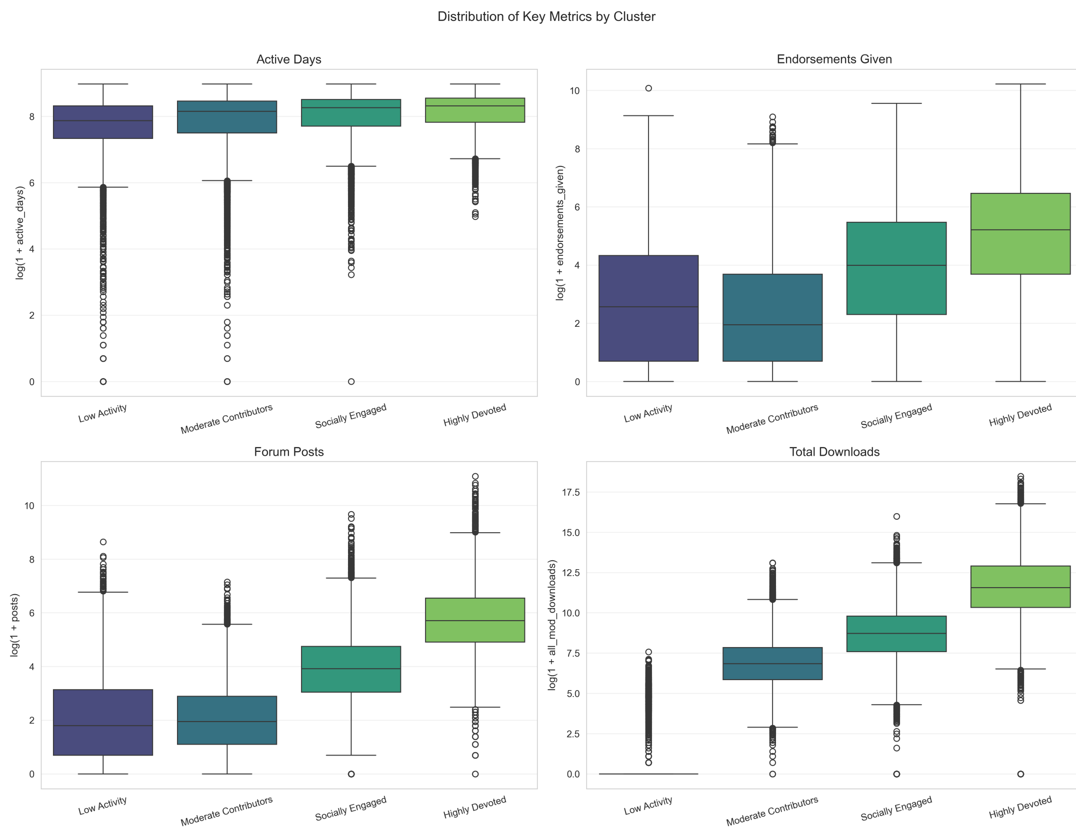


Fig. 7. Boxplot distribution of key metrics across the four modder clusters.

5.8 Temporal Analysis: Time to First Mod

Displays the distribution of "days since joined until first mod upload" for each cluster. The density plots reveal that while all clusters have a peak near zero ("born modders"), the Highly Devoted (Cluster 4) group shows a significantly longer tail and a more spread-out distribution, suggesting many were simple users for years before becoming creators ("made modders").

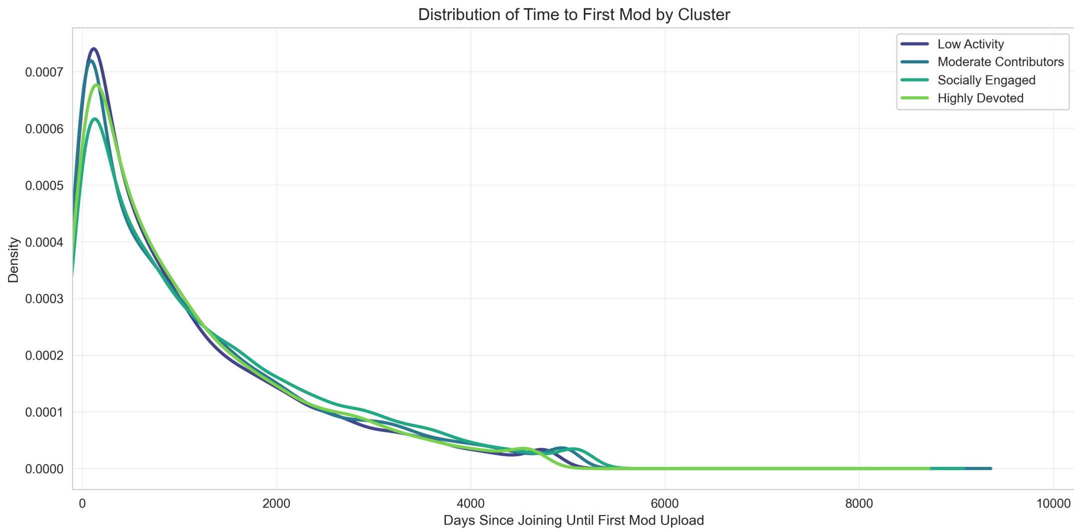
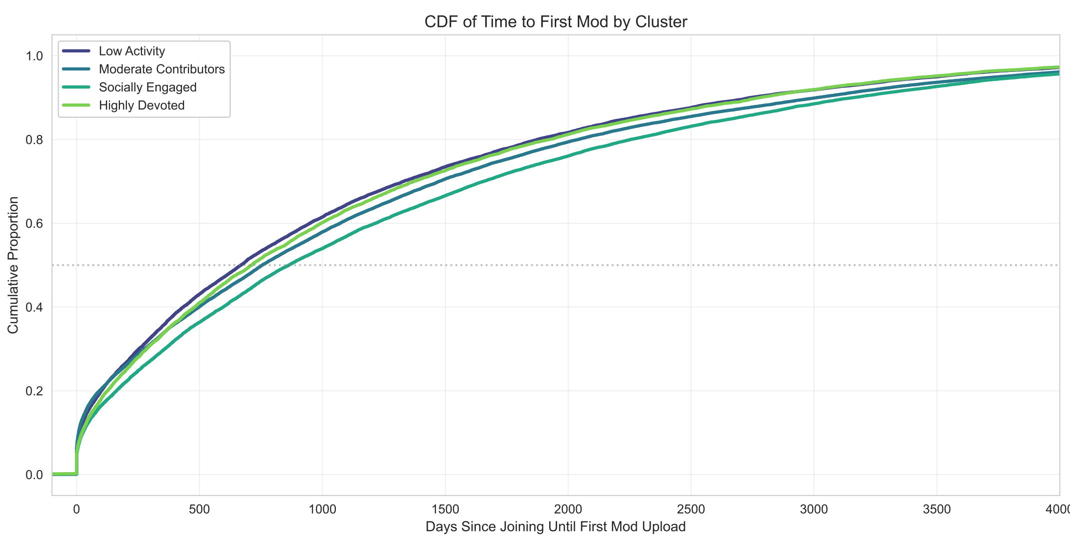


Fig. 8. Distribution of “Time to First Mod” (Days since joining) by Cluster.

5.9 Time to First Mod: CDF

Presents the cumulative distribution function (CDF) of “days from account creation to first mod upload” by cluster, complementing the density plots above. The CDF view highlights that approximately 60% of Low Activity and Moderate users upload their first mod within the first day of joining, whereas only ∼30% of Highly Devoted modders do so—the remainder transition to content creation over months or years.



5.10 Engagement Inequality: Lorenz Curves

Figure 9 displays Lorenz curves for key engagement metrics (downloads, endorsements, forum posts). The pronounced departure from the line of equality confirms that modding engagement follows a highly concentrated distribution: a small percentage of users account for the vast majority of downloads and endorsements, consistent with power-law dynamics observed in other user-generated-content platforms.

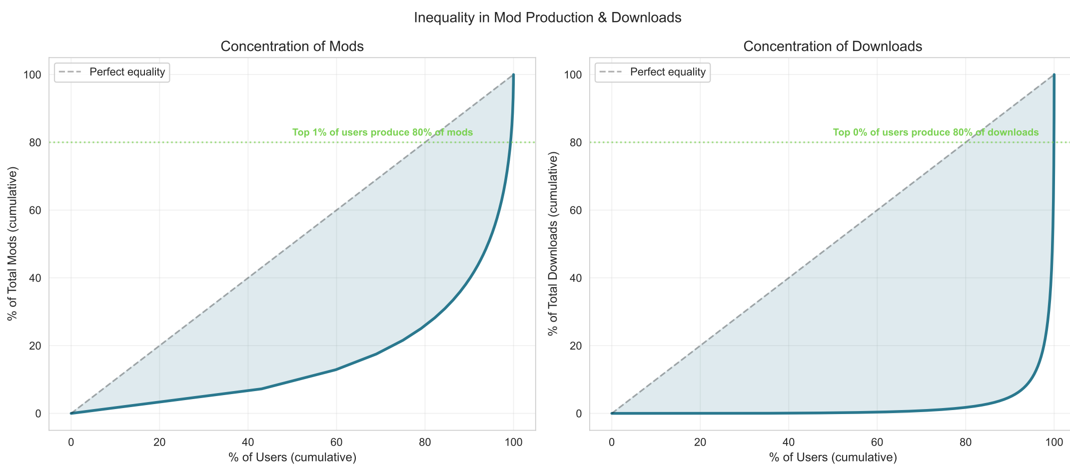


Fig. 9. Lorenz curves for selected engagement metrics, illustrating the concentration of platform activity among a small proportion of users.

5.11 Community Recognition by Cluster

Figure 10 visualises the distribution of community recognition metrics (endorsements received, kudos, page views) across the four modder clusters. The sharp gradient from Cluster 1 to Cluster 4 underscores that platform-visible recognition closely tracks the Devotion Score hierarchy.

Fig. 10. Distribution of community recognition metrics across modder clusters.

5.12 Random Forest Feature Importance

To complement the PCA-derived feature weights (Appendix Table 1), a Random Forest classifier was trained to predict cluster membership from the original (transformed, standardised) features. Figure 11 reports the resulting feature importances, providing a model-agnostic validation of which behavioural metrics most strongly differentiate between modder clusters.

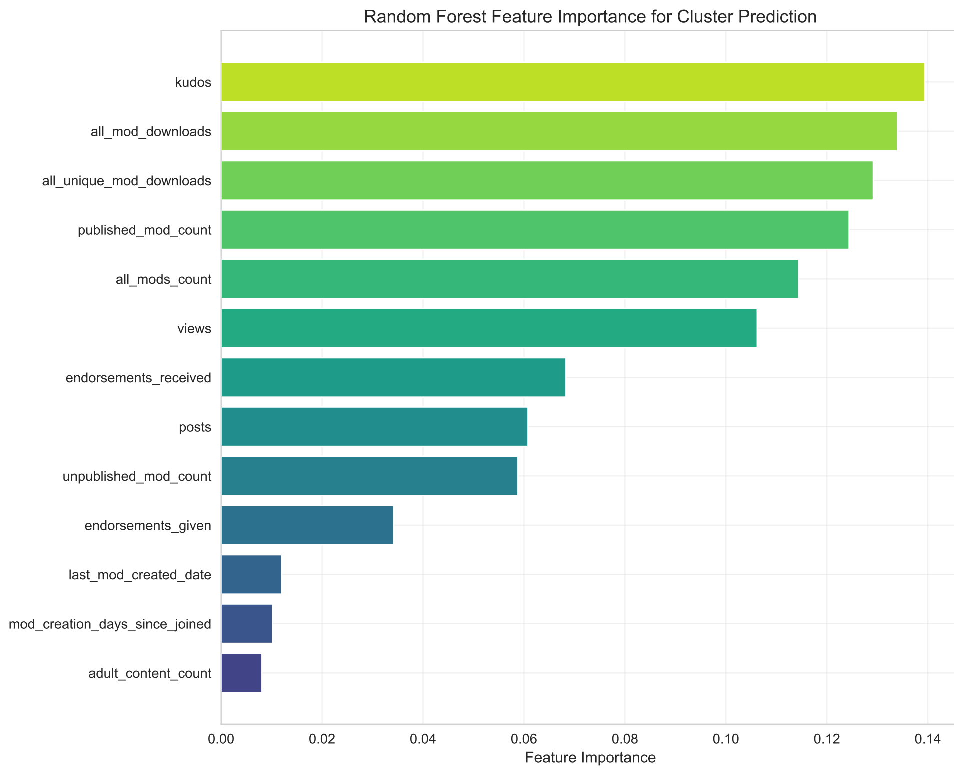


Fig. 11. Random Forest feature importance for predicting modder cluster membership.

5.13 Preprocessing: Per-Feature Transformations

Table 9. Feature-level preprocessing summary. Raw skewness quantifies the original distributional asymmetry; the applied transform was selected to reduce skewness before z-standardisation.

Feature	Raw Skew	Transform	Post Skew
Last mod upload date	-0.90	None	-0.90
Published mod count	22.75	Yeo–Johnson	0.08
Unpublished mod count	36.88	Yeo–Johnson	0.46
Total mods owned	22.04	Yeo–Johnson	0.39
Endorsements given	9.75	$\log(1+x)$	0.21
Forum posts	32.69	$\log(1+x)$	0.29
Kudos	34.39	$\log(1+x)$	1.28
Page views	45.31	Winsorise + $\sqrt[3]{x}$	1.12
Endorsements received	26.12	Yeo–Johnson	0.50
Adult-content count	103.29	Yeo–Johnson	2.18
Total downloads	38.01	Yeo–Johnson	-0.03
Unique downloads	35.42	Yeo–Johnson	-0.03
Days to first mod	1.33	None	1.33